國立臺北科技大學
National Taipei University of Technology

資料結構
Data Structure

Lab 06

姓名:王顥程
學號: 114310215

Lab01-Ex1. (題目編號)
1) Shuffle a standard 52-card deck (Without jokers). **2) Push the shuffled cards into a Queue.** **3) Draw cards from the Queue and start the Blackjack.**
Code

```
#include <iostream>
#include <vector>
#include <string>
#include <algorithm>
#include <random>
#include <ctime>
using namespace std;
struct Player {// 玩家結構體，包含名稱和分數
   string name; // 玩家名稱
   int score;   // 玩家分數
};
class Card {//建立一個Card類別，用來表示單張撲克牌
public:
   string suit; // 花色
   string rank; // 點數
   Card() {}
   Card(string s, string r) : suit(s), rank(r) {}
   void display() const {
      cout << rank << " of " << suit << endl;
   }
};
class Deck {// 牌組類別，負責生成和洗牌
private:
   Card cards[52]; // 52 張牌
   int index; // 目前發牌索引
public:
   Deck() : index(0) {// 初始化牌組
      string suits[] = { "Hearts", "Diamonds", "Clubs", "Spades" };// 四種花色
      string ranks[] = { "2", "3", "4", "5", "6", "7", "8", "9", "10", "J", "Q", "K", "A" };// 13 張牌
      int k = 0;
      for (int i = 0; i < 4; i++) { // 4 種花色
         for (int j = 0; j < 13; j++) { // 13 種點數
            cards[k] = Card(suits[i], ranks[j]);
            k++;
         }
      }
```

```
}
    void shuffleDeck() {// 洗牌
        srand(time(0));
        for (int i = 51; i > 0; i--) {// 從最後一張牌開始，向前遍歷
            int j = rand() % (i + 1);// 生成隨機索引 j
            swap(cards[i], cards[j]);// 交換牌組中的兩張牌
        }
    }
    // 取得指定位置的牌
    Card getCard(int i) const {
        return cards[i];
    }
};
// Queue 類別
class Queue {
private:
    vector<Card> deck; // 用來存放卡牌的容器
    int Front;      // 指向牌堆的前端
    int Rear;       // 指向牌堆的尾端
    int capacity;     // 容器最大容量
public:
    // 建構函式
    Queue(int size) : Front(-1), Rear(-1), capacity(size) {}

    bool isEmpty() const {
        return (Front == -1 || Front > Rear);// 如果 Front 為 -1 或 Front 超過 Rear，代表佇列為
空
    }

    bool isFull() const {
        return (deck.size() >= capacity);// 檢查目前元素數量是否達到上限
    }
    // 插入元素到佇列尾端
    void enqueue(Card card) {
        if (isFull()) { // 檢查是否滿
            cout << "牌堆已滿，無法插入！" << endl;
            return;
        }
        if (Front == -1) {
            Front = 0;
        }
        deck.push_back(card); // 插入元素到尾端
        Rear++;           // 尾端指標移動
    }
```

```
    Card getFront() {// 獲取佇列最前端元素
        if (isEmpty()) { // 檢查是否空
            cout << "牌堆為空，無法獲取最前端的牌！" << endl;
            return Card("", "");
        }
        return deck[Front];
    }
    void dequeue() {
        if (!isEmpty()) {
            Front++; // 移動指標代表移除
        }
    }

};
void initializeDeck(Queue& cardDeck) {// 初始化並洗牌
    Deck deck;
    cout << "初始牌堆: \n";
    for (int i = 0; i < 52; i++) {
        deck.getCard(i).display(); //印出原始牌堆
    }
    deck.shuffleDeck();//洗牌
    cout << "\n洗牌後的牌堆: \n";
    for (int i = 0; i < 52; i++) {
        deck.getCard(i).display();//印出洗牌後的牌堆
    }
    // 將洗好的 52 張牌依序放入 Queue (cardDeck) 中
    for (int i = 0; i < 52; i++) {
        // 呼叫 cardDeck 的 enqueue 方法，將第 i 張牌推入佇列尾端
        cardDeck.enqueue(deck.getCard(i));
    }
}


void initializePlayer(Player* player, string name, Queue& cardDeck) {// 初始化玩家，發兩張牌
    player->name = name;// 玩家名稱
    player->score = 0;// 玩家分數
    cout << player->name << " 抽到的牌: ";

    for (int i = 0; i < 2; i++) {// 發兩張牌
        // 1. 取得佇列最前端的卡牌
        Card drawnCard = cardDeck.getFront();
```

```
            // 2. 將該卡牌從佇列中移除(發牌)
            cardDeck.dequeue();

            // 將抽到的牌印出
            cout << drawnCard.rank << " of " << drawnCard.suit << "  ";

            if (drawnCard.rank == "A") player->score += 1;//計算A分數
            else if (drawnCard.rank == "J" || drawnCard.rank == "Q" || drawnCard.rank == "K")
player->score += 10;//計算JQK分數
            else player->score += stoi(drawnCard.rank);//計算2-10分數
        }
        cout << "\n";
        cout << player->name << " 玩家的初始分數: " << player->score << endl;//印出玩家的
初始分數
    }
    void playerTurn(Player* player, Queue& cardDeck) {// 玩家回合
        char choice;
        while (player->score < 21 && !cardDeck.isEmpty()) {// 玩家小於21點且牌堆不為空
            cout << player->name << " 您的手牌分數目前為: " << player->score << " 要抽牌嗎？
(h = 抽, s = 停) : ";//顯示選擇
            cin >> choice;//輸入選擇

            if (choice == 'h') {//如果玩家選擇抽牌
                // 1. 取得佇列最前端的卡牌並存入 newCard 變數
                Card newCard = cardDeck.getFront();

                // 2. 將該卡牌從佇列中正式移除
                cardDeck.dequeue();
                if (newCard.rank == "A") player->score += 1;//計算A點數
                else if (newCard.rank == "J" || newCard.rank == "Q" || newCard.rank == "K")
player->score += 10; //計算JQK點數
                else player->score += stoi(newCard.rank);//計算2-10點數
                //印出抽到的牌和總點數
                cout << player->name << " 抽到: " << newCard.rank << " of " << newCard.suit << "
總分: " << player->score << endl;
                if (player->score > 21) {//如果玩家大於21點
                    cout << player->name << " 爆掉了！遊戲結束。\n";//玩家爆牌，莊家獲勝
                    return;
                }
            }
            else if (choice == 's') {//如果玩家選擇停牌
                cout << player->name << " 選擇停牌，總分: " << player->score << endl;//印出總點
數
                break;
```

```
            }
            else {
                cout << "請輸入有效選項 (h = 抽, s = 停)！" << endl;//輸入無效
            }
        }
    }
    void dealerTurn(Player* dealer, Queue& cardDeck) {// 莊家回合
        while (dealer->score < 17 && !cardDeck.isEmpty()) {// 莊家小於17點且牌堆不為空
            // 1. 從佇列前端取得一張牌
            Card newCard = cardDeck.getFront();

            // 2. 將該牌從佇列中移除
            cardDeck.dequeue();
            if (newCard.rank == "A") dealer->score += 1;//計算A點數
            else if (newCard.rank == "J" || newCard.rank == "Q" || newCard.rank == "K")
dealer->score += 10;//計算JQK點數
            else dealer->score += stoi(newCard.rank);//計算2-10點數
            //印出抽到的牌和總點數
            cout << "莊家抽到: " << newCard.rank << " of " << newCard.suit << " 莊家目前總分:
" << dealer->score << endl;
            if (dealer->score > 21) {//如果莊家大於21點
                cout << "莊家爆了！玩家獲勝！" << endl;//玩家獲勝
                return;
            }
        }
    }
    // 判斷勝負
    void determineWinner(Player* player, Player* dealer) {
        if (player->score > 21) cout << player->name << " 爆了！莊家獲勝！\n";//玩家爆牌,
莊家獲勝
        else if (dealer->score > 21 || player->score > dealer->score)
            cout << player->name << " 贏了！\n";//莊家爆牌或玩家點數大於莊家, 玩家獲勝
        else if (player->score == dealer->score) cout << "平手！\n";//莊家與玩家點數相同, 平
手
        else cout << "莊家贏了！\n";
    }

    int main() {
        srand(time(0));
        Queue cardDeck(52);//宣告牌堆
        initializeDeck(cardDeck);//初始化牌堆

        Player player, dealer;//宣告並初始化莊家以及玩家
        initializePlayer(&player, "玩家", cardDeck);
```

```
    initializePlayer(&dealer, "莊家", cardDeck);

    playerTurn(&player, cardDeck);
    if (player.score <= 21) {
        cout << "\n莊家回合...\n";
        dealerTurn(&dealer, cardDeck);
        determineWinner(&player, &dealer);
    }

    return 0;
}
```

Lab01-Ex2. (題目編號)

Briefly describe the problem. (題目簡述)

eg. Integrate actual count variable in PrintPairs code.

Code

Paste your code here. Please use Pygments to highlight your code.
(請使用Pygments傳換程式碼格式後再貼上)

Lab01-Ex3. (題目編號)

Briefly describe the problem. (題目簡述)

Code

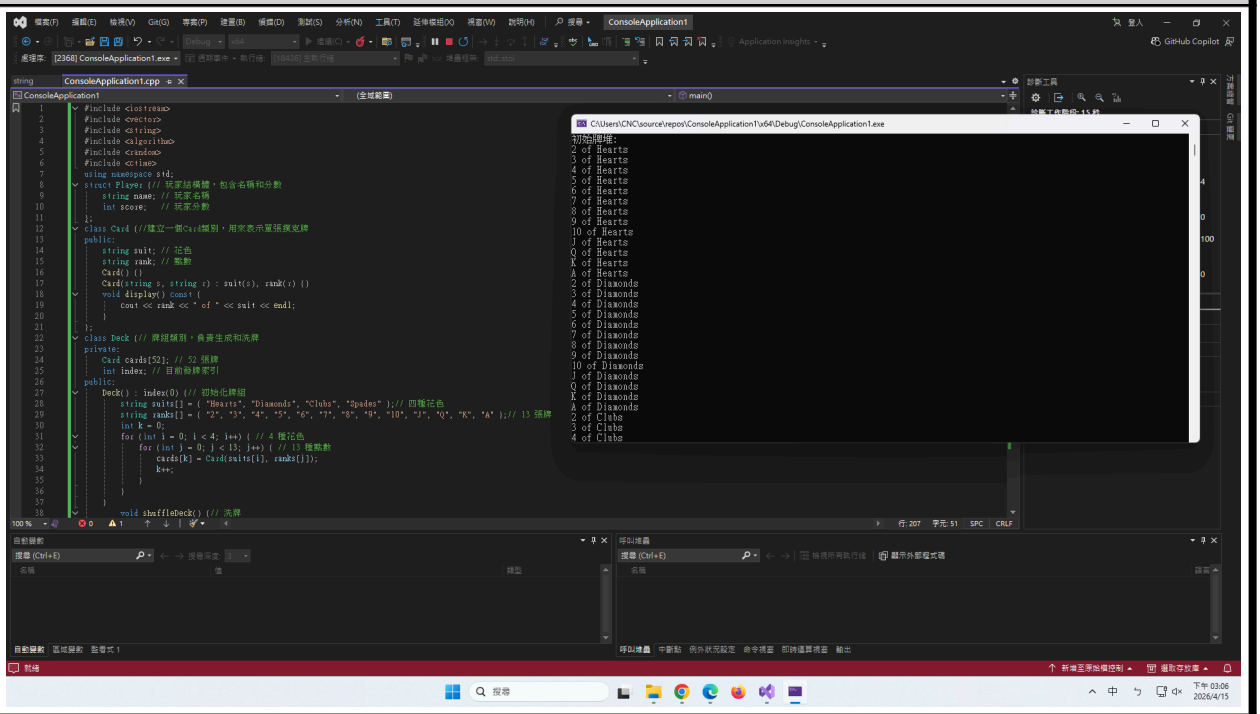
(a)	(b)
Paste your code here. Please use Pygments to highlight your code. (請使用Pygments傳換程式碼格式後再貼上)	Paste your code here. Please use Pygments to highlight your code. (請使用Pygments傳換程式碼格式後再貼上)

Discussion Section
在執行發牌時，雖然兩者都能達成目的，但 Queue 更符合自動化系統的邏輯，而 Stack 則更貼近人類手工發牌的實體感覺。

Lab01-Q1. (題目編號)

Code

Result



Discussion

C:\Users\CNC\source\repos

```
初始牌堆:
2 of Hearts
3 of Hearts
4 of Hearts
5 of Hearts
6 of Hearts
7 of Hearts
8 of Hearts
9 of Hearts
10 of Hearts
J of Hearts
Q of Hearts
K of Hearts
A of Hearts
2 of Diamonds
3 of Diamonds
4 of Diamonds
5 of Diamonds
6 of Diamonds
7 of Diamonds
8 of Diamonds
9 of Diamonds
10 of Diamonds
J of Diamonds
Q of Diamonds
K of Diamonds
A of Diamonds
2 of Clubs
3 of Clubs
4 of Clubs
5 of Clubs
6 of Clubs
7 of Clubs
8 of Clubs
9 of Clubs
10 of Clubs
J of Clubs
Q of Clubs
K of Clubs
A of Clubs
2 of Spades
3 of Spades
4 of Spades
5 of Spades
6 of Spades
7 of Spades
8 of Spades
9 of Spades
10 of Spades
J of Spades
Q of Spades
K of Spades
```

C:\Users\CNC\source\rep

```
洗牌後的牌堆:
6 of Hearts
5 of Diamonds
8 of Clubs
4 of Spades
5 of Hearts
10 of Spades
8 of Diamonds
4 of Clubs
5 of Spades
7 of Hearts
10 of Clubs
K of Diamonds
10 of Diamonds
4 of Diamonds
J of Spades
7 of Clubs
Q of Diamonds
4 of Hearts
7 of Spades
2 of Hearts
9 of Clubs
6 of Clubs
3 of Diamonds
Q of Clubs
J of Diamonds
3 of Spades
J of Hearts
A of Diamonds
2 of Diamonds
J of Clubs
9 of Hearts
9 of Spades
A of Clubs
K of Spades
5 of Clubs
3 of Clubs
6 of Spades
3 of Hearts
10 of Hearts
K of Clubs
K of Hearts
Q of Hearts
7 of Diamonds
2 of Spades
8 of Spades
A of Spades
A of Hearts
2 of Clubs
6 of Diamonds
9 of Diamonds
Q of Spades
```

```
Q of Spades
8 of Hearts
玩家 抽到的牌: 6 of Hearts  5 of Diamonds
玩家 玩家的初始分數: 11
莊家 抽到的牌: 8 of Clubs  4 of Spades
莊家 玩家的初始分數: 12
玩家 您的手牌分數目前為: 11 要抽牌嗎？(h = 抽, s = 停) : h
玩家 抽到: 5 of Hearts 總分: 16
玩家 您的手牌分數目前為: 16 要抽牌嗎？(h = 抽, s = 停) : h
玩家 抽到: 10 of Spades 總分: 26
玩家 爆掉了！遊戲結束。

C:\Users\CNC\source\repos\ConsoleApplication1\x64\Debug\ConsoleApp
若要在偵錯停止時自動關閉主控台，請啟用 [工具] -> [選項] -> [偵錯]
按任意鍵關閉此視窗…
```

Lab01-Q#. (題目編號)		
Briefly describe the problem. (題目簡述)		
Code		
Paste your code here. Please use Pygments to highlight your code. (請使用Pygments傳換程式碼格式後再貼上)	Paste your code here. Please use Pygments to highlight your code. (請使用Pygments傳換程式碼格式後再貼上)	
result		
(a)	(b)	(c)
	Show your result in tables or screenshots. (請使用表格或是截圖呈現結果。)	
Discussion		
Discuss and conclude your results, or answer questions. (實驗討論、結論或回答問題。)		